

MATHEMATICS 2270

Final, Fall semester 2012

Name: Key

1. (10 pts) Find the eigenvalues and the steady state eigenvector of eigenvalue 1 for the Markov matrix $\begin{pmatrix} .6 & .4 \\ .8 & .2 \end{pmatrix}$.

Solution: We have $\det \begin{pmatrix} .6-\lambda & .4 \\ .8 & .2-\lambda \end{pmatrix} = \lambda^2 - .8\lambda - .2 = (\lambda - 1)(\lambda + .2)$ so the eigenvalues are 1 and $-.2$. Set $\lambda = 1$ then $N \begin{pmatrix} -.4 & .4 \\ -.2 & .2 \end{pmatrix}$ is spanned by $(.5, .5)^T$. This is the steady state eigenvector.

2. (10 pts) Find the eigenvalues of $A = \begin{pmatrix} 0 & i & 0 \\ 0 & 0 & 1-i \\ i & 0 & 0 \end{pmatrix}$.

Solution: $\det \begin{pmatrix} -\lambda & i & 0 \\ 0 & -\lambda & 1-i \\ i & 0 & -\lambda \end{pmatrix} = -\lambda^3 - (1-i) = 0$, so $\lambda^3 = i - 1 = 2e^{i3\pi/4}$ and hence $\lambda = 2^{1/3}e^{i(\pi/4+2k\pi/3)}$ where $k = 0, 1, 2$.

3. (10 pts) Find an orthonormal basis for the null space of $\begin{pmatrix} 2 & 1 & -1 & -2 \\ 1 & 0 & 2 & -1 \end{pmatrix}$.

Solution: $N \begin{pmatrix} 2 & 1 & -1 & -2 \\ 1 & 0 & 2 & -1 \end{pmatrix} = N \begin{pmatrix} 0 & 1 & -5 & 0 \\ 1 & 0 & 2 & -1 \end{pmatrix}$ is spanned by $(1, 0, 0, 1)^T$ and $(-2, 5, 1, 0)^T$. To make $(1, 0, 0, 1)^T$ and $(-2, 5, 1, 0)^T$ orthogonal, we replace $(-2, 5, 1, 0)^T$ by $(-2, 5, 1, 0)^T - (1, 0, 0, 1)^T = (-3, 5, 1, -1)^T$. Dividing by the lengths, we get the orthonormal basis

$$2^{-1/2}(1, 0, 0, 1)^T, \quad (28)^{-1/2}(-3, 5, 1, -1)^T.$$

4. (10 pts) Find the Jordan canonical form of $\begin{pmatrix} 2 & t & t+1 \\ 0 & 2 & t+2 \\ 0 & 0 & 2 \end{pmatrix}$. (Note that your answer will depend on t .)

Solution: Let $B = \begin{pmatrix} 0 & t & t+1 \\ 0 & 0 & t+2 \\ 0 & 0 & 0 \end{pmatrix}$, then the rank of B is 2 if $t \notin \{0, -2\}$ and 1 if $t \in \{0, -2\}$.

In the first case the Jordan Canonical Form is $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ and in the second case it is $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

5. (10 pts) Project $v = (1, 1, 2, -3)^T$ on to the subspace spanned by $(1, 1, -1, 2)^T$ and $(2, 1, -1, 1)^T$. (You may use the formula $P = A(A^T A)^{-1}A^T$.)

Solution: Let $A^T = \begin{pmatrix} 1 & 1 & -1 & 2 \\ 2 & 1 & -1 & 1 \end{pmatrix}$, then $A^T A = \begin{pmatrix} 7 & 6 \\ 6 & 7 \end{pmatrix}$ and $(A^T A)^{-1} = \begin{pmatrix} 7/13 & -6/13 \\ -6/13 & 7/13 \end{pmatrix}$. Then $A^T v = (-6, -2)^T$, $(A^T A)^{-1}A^T v = (-30/13, 22/13)^T$ and

$$A(A^T A)^{-1}A^T v = (14/13, -8/13, 8/13, -38/13)^T.$$

6. (10 pts) Find a system of equations whose solutions are given by

$$(1, 1, 1, 1)^T + t(1, 0, 2, -1)^T + s(0, 1, 1, 3)^T.$$

Solution: $N(A) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 1 \\ -1 & 3 \end{pmatrix}$ so that $A = \begin{pmatrix} -2 & -1 & 1 & 0 \\ 1 & -3 & 0 & 1 \end{pmatrix}$. Thus $A(x, y, z, w)^T = (-2, -1)^T$ or $-2x - y + z = -2$ and $x - 3y + w = -1$.

7. (13 pts) a) Define similar matrices $A \sim B$.

Solution: $A \sim B$ if there is an invertible matrix M such that $B = M^{-1}AM$.

b) Show the transitive property; i.e. if $A \sim B$ and $B \sim C$ then $A \sim C$.

Solution: If $A \sim B$ and $B \sim C$ then $B = M^{-1}AM$ and $C = N^{-1}BN$ for invertible matrices M and N . Thus $C = N^{-1}BN = N^{-1}(M^{-1}AM)N = (MN)^{-1}A(MN)$ where MN is invertible i.e. $A \sim C$.

c) Show similar matrices have the same eigenvalues.

Solution: If $A \sim B$ and λ is an eigenvalue of A , then $Av = \lambda v$ for some vector v and $B = M^{-1}AM$. Let $u = M^{-1}v$ then $Bu = M^{-1}AMM^{-1}u = M^{-1}Au = M^{-1}\lambda u = \lambda v$ i.e. v is an eigenvector of eigenvalue λ for B .

8. (12 pts) a) Find a linear map that sends the column vectors of $\begin{pmatrix} 1 & 4 \\ 3 & 1 \end{pmatrix}$ to the column vectors of $\begin{pmatrix} 5 & 2 \\ 4 & 1 \end{pmatrix}$.

Solution:

$$\begin{pmatrix} 5 & 2 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 3 & 1 \end{pmatrix}^{-1} = \frac{-1}{11} \begin{pmatrix} 5 & 2 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & -4 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 1/11 & 18/11 \\ -1/11 & 15/11 \end{pmatrix}.$$

b) Show that there is no linear map sending the column vectors of $\begin{pmatrix} 1 & 4 \\ 2 & 8 \end{pmatrix}$ to the column vectors of $\begin{pmatrix} 2 & 3 \\ 2 & 3 \end{pmatrix}$.

Solution: If $T(1, 2)^T = (2, 2)^T$, then $T(4, 8)^T = 4T(1, 2)^T = 4(2, 2)^T = (8, 8)^T \neq (3, 3)^T$.

9. (15 pts) Solve the system of differential equations $x' = x + 4y$, $y' = 2x + 3y$ with initial value $x(0) = -1$, $y(0) = 3$.

Solution: Let $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. The determinant of $\begin{pmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{pmatrix}$ is $\lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1)$. The eigenvector for $\lambda = 5$ is $(1, 1)^T$ and for $\lambda = -1$ we have $(-2, 1)^T$.

Let $M = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$ and $M^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$. Then $M^{-1}AM = \Lambda$ where $\Lambda = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix}$. If $(u, v)^T = M^{-1}(x, y)^T$ then $(u(0), v(0))^T = M^{-1}(-1, 3)^T = (5/3, 4/3)^T$ and $(u', v')^T = \Lambda(u, v)^T$ so that $(u, v) = (5/3e^{5t}, 4/3e^{-t})^T$. Finally

$$\begin{pmatrix} x \\ y \end{pmatrix} = M \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 5/3e^{5t} \\ 4/3e^{-t} \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 5e^{5t} - 8e^{-t} \\ 5e^{5t} + 4e^{-t} \end{pmatrix}.$$